NAS vs SAN

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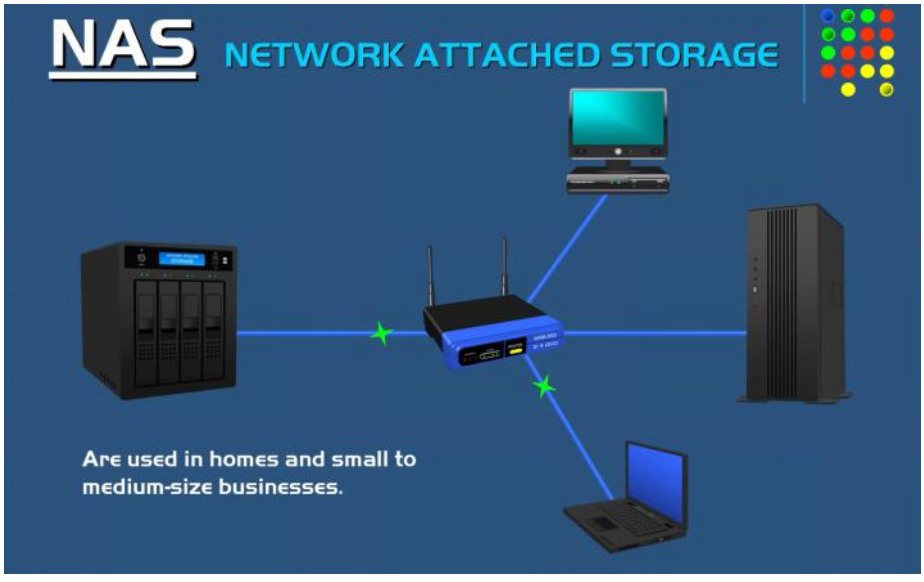
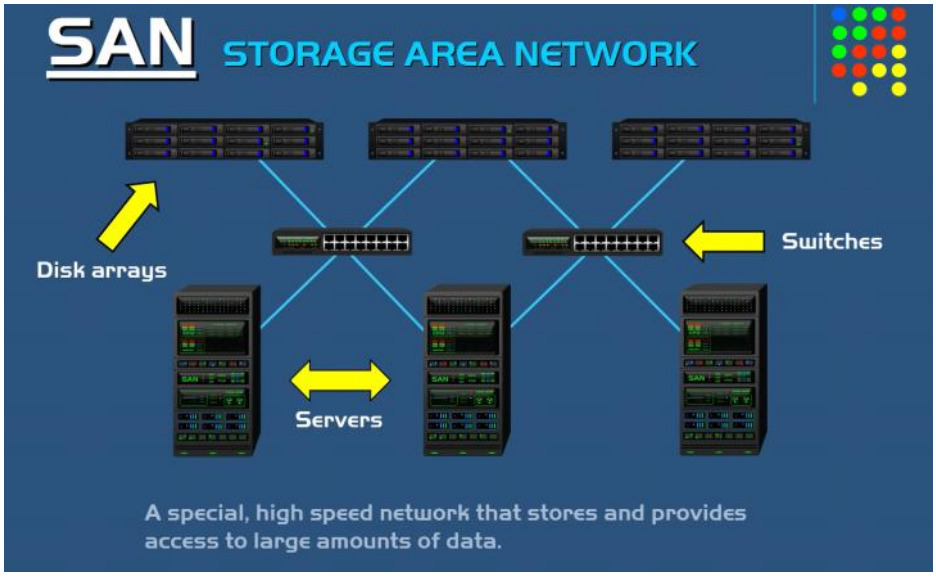
NAS (Network Attached Storage)

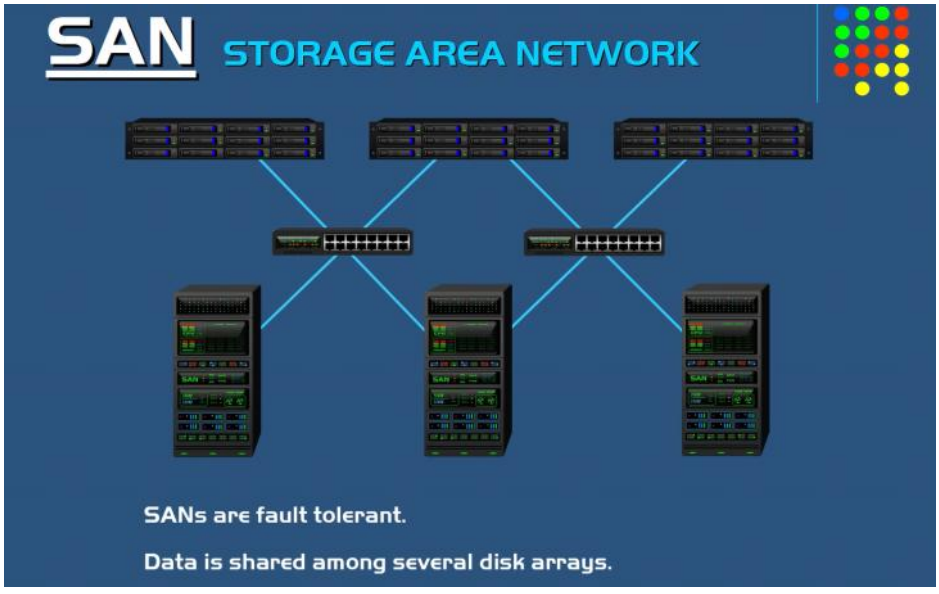
- A storage device connected to a network that lets multiple devices access files as a shared drive.
- Used in homes and small to medium businesses.
- Easy to use and cheaper, but has a single point of failure. Like if the power supply fails on NAS, then the switch cannot access the data on NAS
- IT also have NIC which can be connected to switch or router



SAN (Storage Area Network)

- A high speed dedicated network for storage with multiple devices like disk arrays and servers.
- Data is accessed like a local hard drive, not a shared drive.
- Very fast, scalable, and fault tolerant, but expensive and mainly used by large organizations. Can be connected with fibre cable for complete performance.





Key Difference :

NAS = simple file sharing over a network
SAN = high performance, dedicated storage network for large scale use